

Labwork 3

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1 Introduction

In this labwork, we implement the logistic regression algorithm using gradient descent to find the best-fitting line for a given dataset. We will analyze the convergence of the algorithm and observe how different learning rates affect the results.

2 Implementation and Experimentation

The dataset used in this labwork is `loan2.csv`, which consists of samples with two input features — **Experience** ($x^{(1)}$) and **Salary** ($x^{(2)}$) — and a binary target label $y \in \{0, 1\}$ indicating whether a loan was approved. The goal of logistic regression is to find the parameters w_1 , w_2 , and w_0 such that the model

$$\hat{y} = \sigma(w_1x^{(1)} + w_2x^{(2)} + w_0)$$

best separates the two classes, where σ is the sigmoid function defined as

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

This means we want to find (w_1, w_2, w_0) that minimizes the loss function:

$$J = -\frac{1}{N} \sum_{i=1}^N \left(y_i(w_1x_i^{(1)} + w_2x_i^{(2)} + w_0) - \log \left(1 + e^{w_1x_i^{(1)} + w_2x_i^{(2)} + w_0} \right) \right)$$

where N is the number of data points.

All the related functions and the gradient descent algorithm are implemented in the Python file `labwork2_logistic.ipynb`. The main components of the implementation include:

- The loss function $J(w_1, w_2, w_0)$ which we want to minimize, defined as:

$$J = -\frac{1}{N} \sum_{i=1}^N \left(y_i(w_1x_i^{(1)} + w_2x_i^{(2)} + w_0) - \log \left(1 + e^{w_1x_i^{(1)} + w_2x_i^{(2)} + w_0} \right) \right)$$

- The derivatives of the loss function with respect to w_1 , w_2 , and w_0 , computed using the formulas:

$$\frac{\partial J}{\partial w_0} = \frac{1}{N} \sum_{i=1}^N \left(-y_i + \left(1 - \sigma(-(w_1 x_i^{(1)} + w_2 x_i^{(2)} + w_0)) \right) \right)$$

$$\frac{\partial J}{\partial w_1} = \frac{1}{N} \sum_{i=1}^N \left(-y_i x_i^{(1)} + x_i^{(1)} \left(1 - \sigma(-(w_1 x_i^{(1)} + w_2 x_i^{(2)} + w_0)) \right) \right)$$

$$\frac{\partial J}{\partial w_2} = \frac{1}{N} \sum_{i=1}^N \left(-y_i x_i^{(2)} + x_i^{(2)} \left(1 - \sigma(-(w_1 x_i^{(1)} + w_2 x_i^{(2)} + w_0)) \right) \right)$$

- The gradient descent algorithm that updates the values of w_1 , w_2 , and w_0 at each iteration based on the computed gradients and a specified learning rate ℓ :

$$w_k \leftarrow w_k - \ell \cdot \frac{\partial J}{\partial w_k}, \quad k \in \{0, 1, 2\}$$

The update stops when the change in the loss function between two consecutive iterations is less than a specified threshold $\varepsilon = 10^{-6}$, indicating that the algorithm has converged to a minimum.

3 Results and Discussion

With this problem, we set the initial values of w_1 , w_2 , and w_0 to 1, 0, and 0 respectively. The initial learning rate is set at 0.001. The algorithm converge after over 96000 steps, with the final results being $w_1 \approx 2.16$, $w_2 \approx 0.14$, and $w_0 \approx -3.15$. The fitted regression line is plotted in the figure below, showing how the model separates the data points based on the learned parameters.

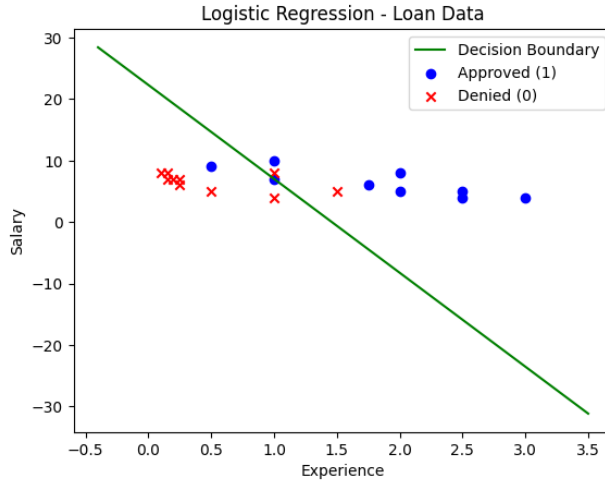


Figure 1: Data points and the fitted regression line

With a lower learning rate of 0.0001, unlike the previous two labworks, the algorithm converges much faster after only 7107 steps, with the final parameters being $w_1 \approx 0.92$, $w_2 \approx -0.25$, and $w_0 \approx 0.79$. The fitted regression line is plotted in the figure below, showing a different separation of the data points compared to the previous case, which shows that the choice of learning rate can significantly affect the convergence and the final model parameters.

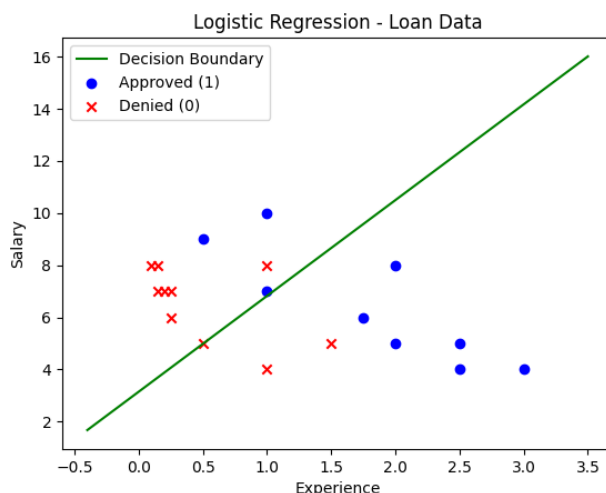


Figure 2: Data points and the fitted regression line with a lower learning rate

Finally, with a higher learning rate of 0.1, the algorithm converges after over 13000 steps, with the final parameters being $w_1 \approx 4.54$, $w_2 \approx 1.27$, and $w_0 \approx -13.57$. The fitted regression line is plotted in the figure below, showing the model's performance with the higher learning rate.

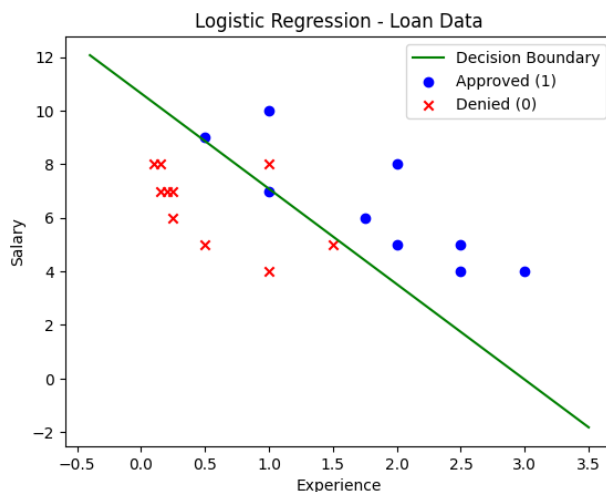


Figure 3: Data points and the fitted regression line with a higher learning rate